

From Regression to Randomization:
Estimating the Causal Effect of Health Insurance

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10616720: Statistical Inference

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Abstract

This paper asks whether health insurance causally improves health and financial well-being, and it uses the contrast between two research designs to show how much the answer depends on identification. Part I regresses state-level mortality on the insured share in a 2019 U.S. cross-section ($n = 51$). The estimated association is small, unstable across specifications, and wrongly signed in the preferred age-adjusted model; we argue it cannot be read causally, because omitted variables, reverse causality, and the ecological fallacy confound any aggregate observational regression. Part II replicates the Oregon Health Insurance Experiment (Finkelstein et al., 2012), in which the right to apply for Medicaid was assigned by lottery. The first stage is strong ($\approx +0.26$), pre-randomisation covariates are balanced, and survey non-response is small. Medicaid increased healthcare use, substantially reduced financial strain (the most robust result), and improved self-reported and mental health; effects on physical biomarkers are null and the full-sample mortality effect is a precise, underpowered null. Estimating the same enrolment effect by OLS and matching yields near-zero coefficients far from the experimental LATE, demonstrating selection on unobservables directly. The methodological thesis is that credible causal knowledge comes from identification, not from statistical adjustment.

Keywords: Medicaid; causal inference; randomized experiment; ITT; LATE; selection bias; omitted-variable bias.

Part I: Cross-Sectional Analysis Using Regression

1. Data Exploration

We use a cross-section of the 50 U.S. states plus the District of Columbia for 2019 ($n = 51$). The dependent variable is the state mortality rate (deaths per 100,000), available both as a crude rate and as a CDC age-adjusted rate (direct standardisation to the 2000 U.S. Standard Population). The main regressor is the insured share of the population (%); controls are the median age, the poverty rate (%), and log median household income. Insurance and poverty, originally per-100,000, are rescaled to percentages so that coefficients read as the effect of a one-percentage-point change.

Table 1

Summary statistics ($n = 51$)

Variable	Mean	Median	Min	Max
Crude mortality (per 100k)	911.0	930.5	584.4	1,305.9
Insured share (%)	91.6	92.0	81.6	97.0
Poverty rate (%)	12.2	11.8	7.3	19.6
Median age (years)	38.8	38.8	31.2	45.1
Log median income	11.08	11.05	10.73	11.43

Two features frame the entire analysis. First, the insured share is right-skewed and compressed: the median state already insures 92% of its population, so the identifying variation in coverage is thin. Second, the bivariate correlations already expose the confounding (Figure 1): mortality is strongly associated with income, median age, and poverty, yet almost uncorrelated with the insured share itself (+0.06). Crucially, the insured share is in turn correlated with income and median age, the two strongest correlates of mortality, which is precisely the configuration under which an omitted control distorts the coefficient of interest.

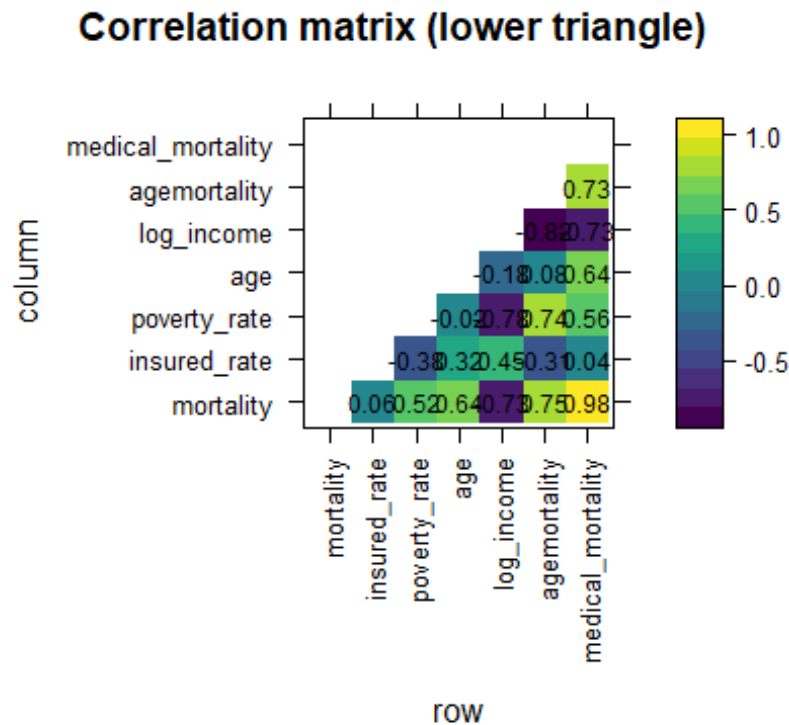


Figure 1. Pairwise correlation matrix of the main variables (lower triangle). The insured share is essentially uncorrelated with mortality (+0.06) but is itself correlated with income and median age, which are in turn the strongest correlates of mortality, the configuration that generates omitted-variable bias in a simple bivariate regression

A single high-mortality state is a visible outlier; re-estimating the bivariate model after excluding states above 1,300 deaths per 100,000 leaves the insurance coefficient statistically indistinguishable from zero. The near-zero raw association is therefore a genuine data feature, not a leverage artefact.

2. Motivation

The research question is whether a higher share of insured population causally lowers state mortality. The theoretical prior is unambiguous: insurance lowers the price of care, raises utilisation, and should reduce avoidable deaths, implying a negative coefficient on coverage (hypothesis H1: $\beta_{\text{insurance}} < 0$). The empirical aim of this section, however, is deliberately diagnostic: we show that a purely observational, aggregate design cannot recover this causal parameter, regardless of how carefully the regression is specified, thereby motivating the experimental strategy of Part II.

3. Model Specification

We estimate a sequence of nested OLS specifications. The continuous dependent variable is mortality (crude in Models 1–4, age-adjusted in the preferred Model 5); the main explanatory variable is the insured share; controls enter progressively. Expected signs are: median age (+), poverty (+), income (–). To keep main effects interpretable in the presence of an interaction, the insured share and log income are mean-centered in Models 4–5, so each main effect is evaluated at the sample mean of the other.

Two specification choices matter. First, we reject a quadratic in median age: its terms are individually insignificant yet imply economically implausible marginal effects ($\approx +63$ deaths per 100k at the youngest state median), an artefact of collapsing a sharply nonlinear individual age–mortality relationship onto a compressed range of state medians (~ 31 – 45 years). Second, and consequently, our preferred outcome is age-adjusted mortality: standardising at the construction stage removes the age-composition channel (including the mechanical Medicare link, whereby older states show both near-universal coverage and high crude mortality) far more cleanly than a noisy age polynomial on the right-hand side. Finally, given $n = 51$ and a few high-leverage states, we report HC3 heteroskedasticity-robust standard errors for inference on the preferred model (MacKinnon and White, 1985).

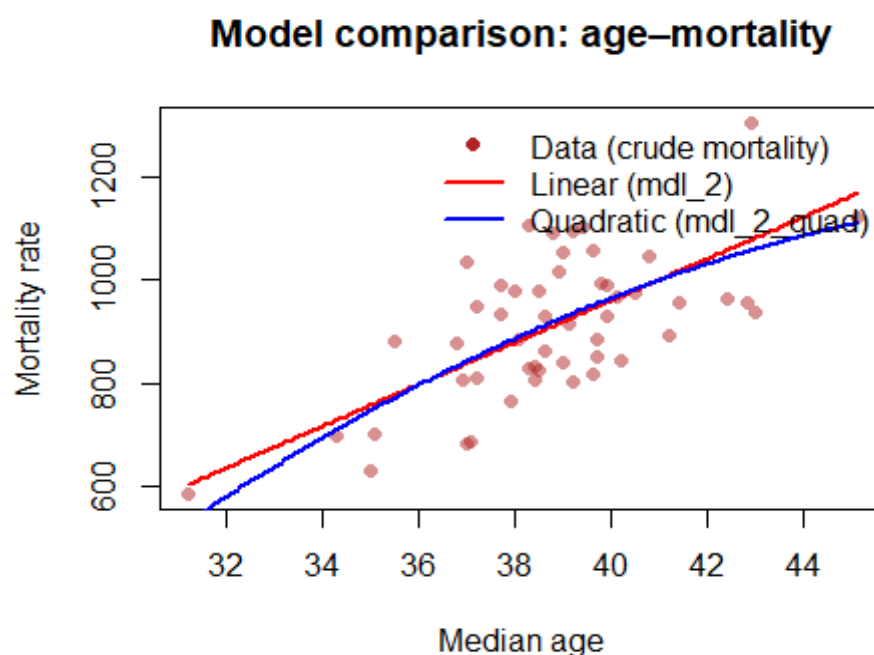


Figure 2. Median age versus mortality with the linear and quadratic fits. The quadratic curvature is an aggregation artefact, motivating age-adjustment of the outcome rather than an age polynomial.

4. Regression Analysis

Table 2
OLS estimates of mortality on the insured share

	(1)	(2)	(3)	(4)	(5)
Dependent variable	Crude	Crude	Crude	Crude	Age-adj.
Insured share (%)	2.63 (6.36)	-7.12 (5.11)	12.34*** (3.09)	12.61*** (2.99)	2.25 (2.53)
Median age		40.77*** (6.75)	24.94*** (3.84)	23.87*** (3.75)	
Poverty rate (%)			2.39 (4.77)	-3.37 (5.41)	7.14 (4.91)
Log median income			-618.42*** (84.25)	-724.18*** (96.51)	-360.03*** (82.39)
Insured $\times$ Log income				39.73** (19.43)	9.62 (18.25)
Constant	670.12 (583.20)	-18.72 (458.63)	5633.64*** (963.78)	16.90 (174.90)	652.38*** (58.11)
Observations	51	51	51	51	51
R ²	0.003	0.434	0.859	0.871	0.699
Adjusted R ²	-0.017	0.410	0.847	0.857	0.672

Notes: OLS coefficients; classical standard errors in parentheses. Models (4)–(5) use mean-centered insured share and log income. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Model (5) is the preferred specification; under HC3 robust SEs the insured-share coefficient is 2.25 (SE 2.79, $p = 0.42$), log income -360.03 (SE 99.57, $p < 0.01$), and the interaction 9.62 (SE 21.64, $p = 0.66$): every qualitative conclusion is unchanged. Robust joint tests on Model (5) confirm that the insurance bundle (direct effect + interaction) is jointly insignificant ($F = 0.58$, $p = 0.56$), whereas the income bundle (income + poverty) is highly significant ($F = 35.0$, $p < 0.01$).

The sign of the insurance coefficient is entirely an artefact of conditioning, and tracking it across columns is the substantive content of the table. In the bivariate Model 1 the estimate is positive, tiny, and insignificant (+2.63), the opposite of the protective sign theory predicts. Adding median age (Model 2) flips it negative (-7.12) but still insignificant: omitting age had biased the coefficient upward through the age–Medicare channel. Adding income and poverty (Model 3) reverses it again to a large, significant positive value (+12.34), a textbook suppression pattern: income is strongly protective and positively correlated with coverage, so omitting it had masked a positive conditional association. Model 4 confirms this on crude mortality, with a marginally significant insurance $\times$ income interaction. In the preferred Model 5 (age-adjusted outcome, income interaction) the coefficient collapses back toward zero

(+2.25, $p = 0.42$ robust): once age composition is removed at the source, no protective association survives, and the point estimate even carries the wrong (positive) sign.

The R^2 progression (0.003 $\rightarrow$ 0.86 across Models 1–4) confirms that omitted-variable bias dominated the bivariate model, but it is not a model-selection criterion: it never falls when regressors are added, and Model 5's lower R^2 (0.70) is not comparable to Models 1–4 because it has a different dependent variable. Three further interaction specifications (insurance $\times$ poverty, $\times$ income, $\times$ age) were estimated on the age-adjusted outcome; none of the insurance moderations is significant (all interaction $p > 0.4$), so we find no evidence that the coverage–mortality association varies systematically with state poverty, income, or age structure.

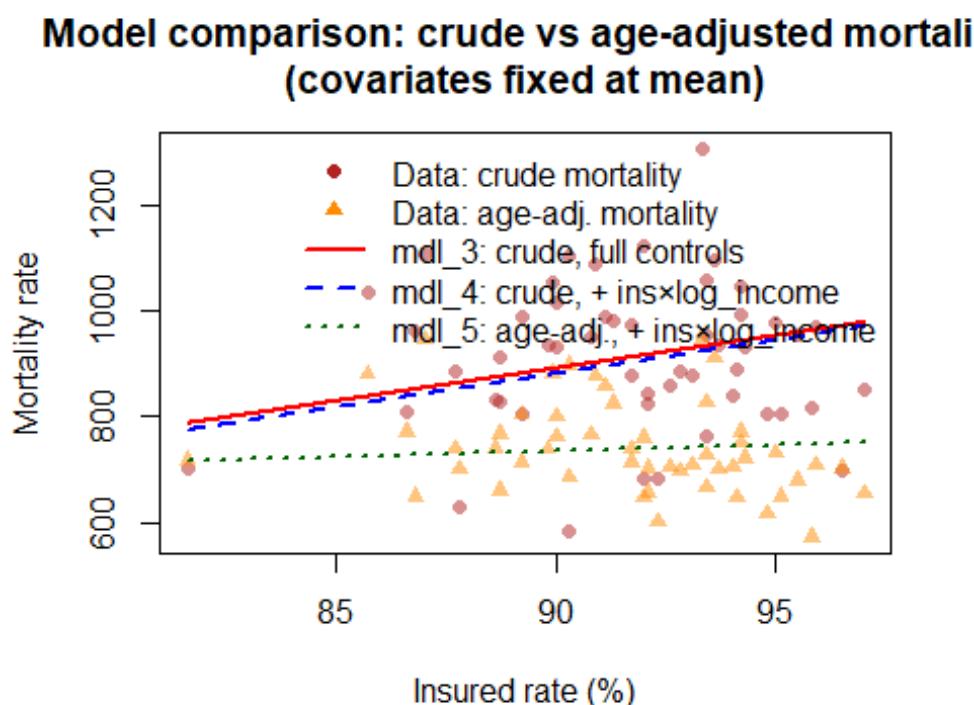


Figure 3. Fitted state-level relationship between the insured share and mortality. Age-adjusting the outcome (Model 5) flattens the steep, spurious positive slope of the crude-mortality models toward zero, but does not produce the protective (negative) association predicted by theory.

Robustness: amenable (medical) mortality

As a final check we re-estimate the preferred specification on age-adjusted amenable (medical) mortality, deaths from causes most plausibly responsive to medical care, which should give coverage its best chance of showing a protective association (Table 3). It does not: the insured-share coefficient stays small, positive, and statistically indistinguishable from zero, while income retains its strong protective coefficient. The diagnostic conclusion is therefore not an artefact of using all-cause mortality.

Table 3

Robustness: insured-share effect on all-cause versus amenable mortality

	All-cause (Model 5)	Amenable (medical)
Insured share (%)	2.25 (2.53)	1.37 (1.67)
Log median income	−360.03*** (82.39)	−199.09*** (54.15)
Insured × Log income	9.62 (18.25)	6.84 (11.99)
Poverty rate (%)	7.14 (4.91)	7.78* (3.22)
Observations	51	51
R ²	0.699	0.711

Note. Dependent variable: age-adjusted all-cause versus age-adjusted amenable (medical) mortality; classical SEs in parentheses; insured share and log income mean-centered. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5. Discussion and Validity

The central result is that even the most defensible cross-sectional specification cannot recover a credible causal effect of coverage on mortality: the coefficient is statistically zero and wrongly signed, and the apparent instability across columns is driven by confounding, not by a true economic relationship. This is a diagnosis, and four threats to internal validity make a causal reading untenable within this design:

- **Omitted-variable bias.** The engine of the unstable signs. Beyond the age and income channels we can observe, the model omits racial composition, urbanisation, health behaviours, opioid prevalence, and healthcare-system quality, all plausibly correlated with both coverage and mortality.
- **Reverse causality.** Causation may run the other way: states with sicker or poorer populations expand Medicaid in response to high mortality, attenuating or reversing the coefficient.
- **Measurement error.** The insured share captures any coverage, not its generosity: a state where 92% hold catastrophic-only plans is treated identically to one where 92% hold comprehensive plans.
- **Ecological fallacy / aggregation bias.** Relationships estimated on state aggregates need not hold at the individual level (Deaton and Lubotsky, 2003). Age-adjusting the outcome addresses one facet of this problem, but the design remains irreducibly ecological.

It is worth being explicit about why the sign of the coverage coefficient moves around so much, because the mechanism is instructive rather than anomalous. Two confounding channels operate in opposite directions at once. The first is a mechanical age–Medicare channel: older states have both higher mortality and near-universal coverage through Medicare, so failing to hold age composition constant biases the insurance coefficient upward (toward appearing harmful), which is exactly why the estimate turns negative the moment median age enters in Model 2.

The second is income acting as a suppressor: richer states have lower mortality and higher coverage, so omitting income lets the insurance variable proxy for income's protective effect and masks a positive conditional association, which is why the coefficient jumps strongly positive once income enters in Model 3. These two biases partially cancel in the bivariate model, manufacturing the near-zero estimate a careless reader might mistake for a finding; the progressive specification is what unmask the cancellation.

On external validity, even an internally valid estimate would generalise poorly: the analysis is a single 2019 U.S. snapshot, coverage is bunched near a ceiling that leaves little informative variation, and the estimand is a between-state association rather than an individual treatment effect.

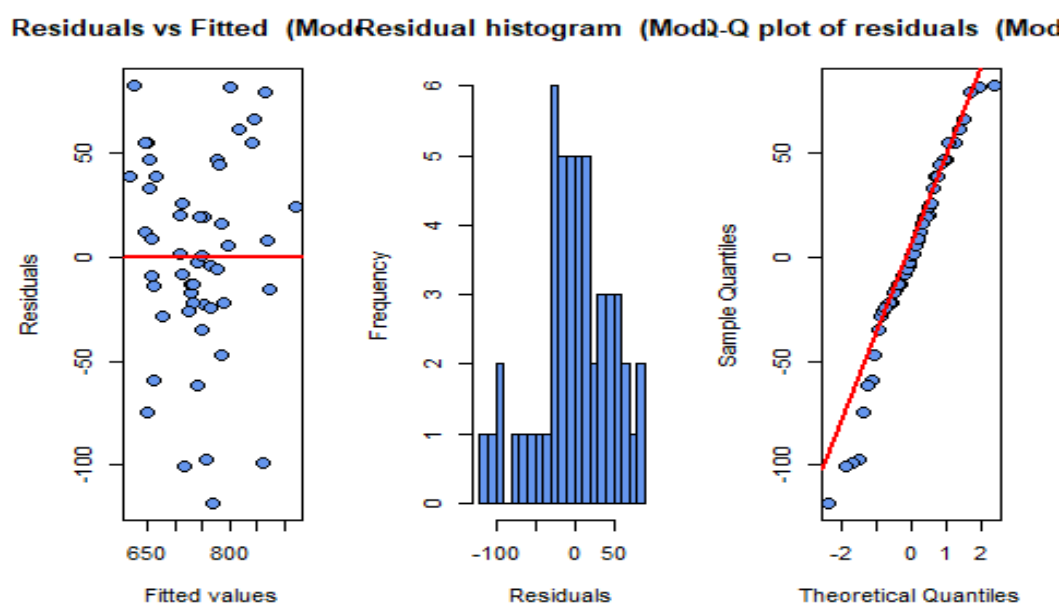


Figure 4. Residual diagnostics for the preferred model (residuals versus fitted values and a normal Q–Q plot). With $n = 51$ these checks are read jointly and cannot resolve the underlying identification problem.

The residual diagnostics for the preferred model (Figure 4) are consistent with this reading. The residuals-versus-fitted plot shows no gross functional-form pathology and only mild non-constant spread, which is why we report heteroskedasticity-robust standard errors; the histogram and normal Q–Q plot are roughly bell-shaped, with the modest tail deviation expected at $n = 51$. Crucially, well-behaved residuals do not vindicate the specification: if important confounders are omitted, their variation is simply absorbed into the error term, leaving the residuals looking innocuous while the model remains misspecified in the causal sense that matters here.

The unavoidable conclusion is that identification, not additional statistical adjustment, is what a causal estimate requires, and that is exactly what a cross-section of observational, aggregate data cannot supply. Quasi-experimental designs that exploit plausibly exogenous variation in coverage do find mortality effects (e.g. Sommers, Baicker and Epstein, 2012). Part II turns to one such source: the 2008 Oregon Health Insurance Experiment, in which the right to apply for Medicaid was allocated by lottery. Randomisation severs the link between coverage and the unobserved determinants of health that contaminate every column of Table 2, so a simple comparison of lottery winners and losers admits a causal reading. Mortality, the outcome that anchors Part I, reappears there as the shared bridge between the two designs.

Part II: The Causal Effects of Medicaid Coverage:

Evidence from the Oregon Health Insurance Experiment

1. Research Question and Treatment Definition

We ask whether Medicaid coverage causally improves healthcare use, financial security, and health, and whether, as the outcome shared with Part I, it reduces mortality. The setting is the 2008 Oregon Health Insurance Experiment (OHIE): facing more eligible low-income adults than its budget could cover, Oregon allocated the right to apply for Medicaid by random lottery, creating a genuine randomized controlled trial embedded in policy (Finkelstein et al., 2012).

The analysis turns on a distinction that organises the entire section. The lottery randomises an offer of coverage, not coverage itself, so two treatment variables and two estimands must be kept separate:

- the offer (winning the lottery), whose effect is the intention-to-treat (ITT), identified by randomisation alone;
- actual enrolment in Medicaid, whose effect for the subpopulation induced to enrol by winning is the local average treatment effect (LATE), identified by instrumenting enrolment with the lottery.

The control group is the lottery losers. Outcomes span three OHIE domains (healthcare use, financial strain, and self-reported/mental health), each summarised by a standardised index, plus administrative mortality (observed for everyone). The problem is causal precisely because enrolment is endogenous: people who take up Medicaid differ from those who do not in health, expected need, and motivation: selection on unobservables. This is exactly the pathology Part I could not escape; the lottery is what breaks it.

The analysis uses the Oregon Health Insurance Experiment public-use files (Harvard Dataverse). The full lottery sample contains 74,922 individuals, for whom the offer, household identifiers, and administrative outcomes (Medicaid enrolment and mortality) are observed; the survey-based outcomes come from the twelve-month mail survey, restricted to responders with a non-zero survey weight (roughly twenty-four thousand individuals). We keep the two samples distinct throughout: administrative outcomes are estimated on the full sample, survey outcomes on the weighted responder sample.

2. Identification Strategy

The identification challenge is that a naive comparison of the insured and the uninsured confounds the effect of coverage with the selection that determines who is covered. Randomisation of the offer solves this for the ITT: because winners and losers are comparable in expectation, the difference in their mean outcomes is the causal effect of the offer. The only qualification is that the lottery gave larger households a higher probability of winning, so randomisation holds conditional on household size; every specification therefore includes household-size fixed effects.

The LATE requires the standard instrumental-variable assumptions: relevance (the offer must move enrolment), independence (random assignment), exclusion (winning affects outcomes only through enrolment, which is plausible since the lottery changed nothing but the right to apply), and monotonicity (no one enrolls because they lost). We defend these assumptions empirically with three design checks, summarised in Table 4.

Table 4
Research-design validation

Check	Tests assumption	Estimate (clustered SE)	Reading
First stage	Relevance	0.256*** (0.0035)	Winning raises “ever enrolled” by ≈ 25.6 pp
Balance (omnibus F, 8 covariates)	Random assignment	F = 1.31, p = 0.23	Cannot reject balance
Differential non-response	Survey validity	-0.017*** (0.0044)	Winners respond ≈ 1.7 pp less

Note. household-clustered SEs. The first stage regresses administrative enrolment on the offer with draw and household-size fixed effects; the balance F-test regresses the offer on all pre-randomisation covariates jointly. The single covariate with a large unconditional imbalance (self-enrolment) is mechanically tied to household size and is balanced once conditioned on it (diff +0.0003, p = 0.27).

The first stage is strong, so the offer is a real treatment and the instrument is not weak. Balance confirms that randomisation worked. The one genuine wrinkle is differential non-response: winners are slightly less likely to return the voluntary 12-month survey. The gap is tiny (≈ 1.7 pp; response rates $\approx 41.5\%$ versus $\approx 39.9\%$) and significant only because the sample is enormous, but it is real, can re-introduce selection among respondents, and is therefore carried forward as a limitation rather than dismissed. All survey estimates use the OHIE design/non-response weights.

3. Estimation

Estimation uses weighted OLS for the ITT (with survey-wave $\times$ household-size fixed effects), two-stage least squares for the LATE, and propensity-score matching as an observational benchmark. Inference relies on household-clustered standard errors throughout, since treatment is assigned at the household level. Three steps go beyond the course labs and are flagged with their sources: clustered SEs (Cameron and Miller, 2015; Zeileis, Köll and Graham, 2020), IV/2SLS (Imbens and Angrist, 1994; Angrist and Pischke, 2009), and randomisation inference (Imbens and Rubin, 2015); all other commands are course-native.

3.1 Selection bias: observational versus experimental (the bridge to Part I)

On the same OHIE sample we estimate the effect of enrollment three ways and watch the observational answer fail (Table 5). A crucial estimand note: OLS, matching, and LATE all target the effect of enrolment, whereas the ITT targets the effect of the offer; the valid selection-bias contrast is therefore OLS/matching versus LATE, never OLS versus ITT.

Table 5

Effect of Medicaid enrolment: observational versus experimental estimators

Outcome	OLS (observational)	Matching, ATT (observational)	LATE / 2SLS (experimental)
Good/very-good health	-0.008 (0.007)	-0.010 (0.009)	0.133*** (0.026)
Owe medical debt	-0.062*** (0.008)	-0.066*** (0.009)	n/a

Notes: binary outcomes; household-clustered SEs. The ITT is deliberately excluded (different estimand). LATE was estimated only for health. For health, OLS and matching agree with each other (≈ 0) but lie far outside the confidence interval of the experimental LATE (+0.13): the gap is selection bias, not noise. One-to-one nearest-neighbour propensity-score matching balances every observed covariate (standardised differences fall from ≈ 0.22 to inside the 0.10 band; see balance figure) yet still misses the experimental answer, because enrolment is selected on unobservables that no observable adjustment can see.

This is Part I's diagnosis reproduced in miniature and then resolved: controlling for more covariates does not buy identification; only exogenous variation (here, a literal lottery) does.

Covariate Balance Before/After NN Matching (Medicaid)

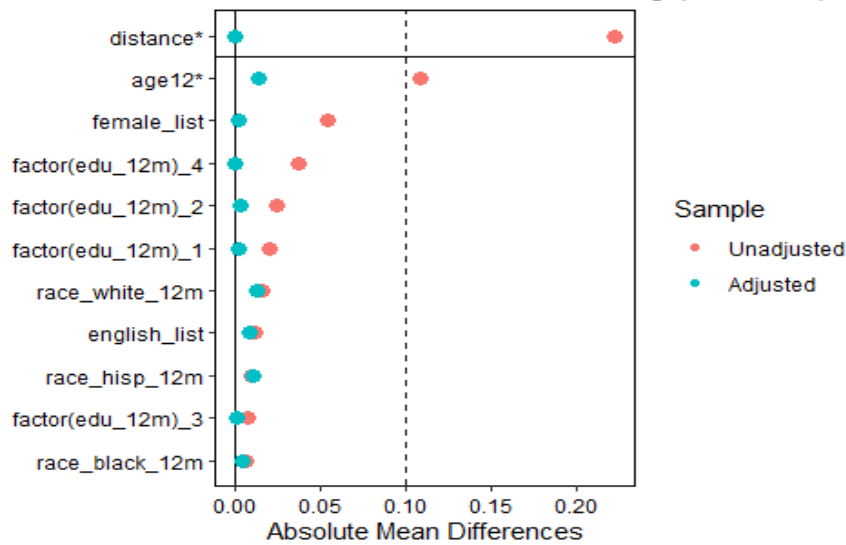


Figure 5. Covariate balance before and after 1:1 nearest-neighbour propensity-score matching (standardised mean differences; dashed lines at ± 0.10). Matching removes imbalance on all observed covariates.

Effect of Medicaid ENROLLMENT: observatic OLS & Matching (blind to unobservables) vs LATE (lotter

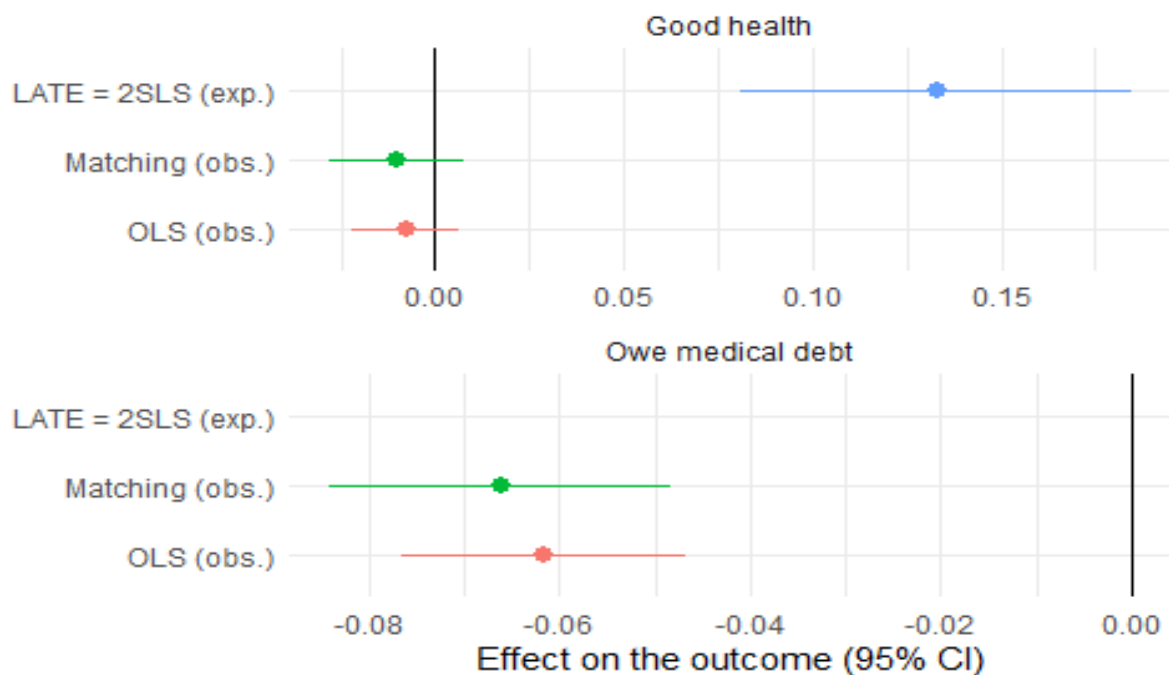


Figure 6. Effect of Medicaid enrolment estimated by OLS and matching (observational) versus LATE/2SLS (experimental), with 95% confidence intervals. The ITT is omitted because it targets a different estimand.

3.2 Main causal effects: ITT across the three domains

We report the ITT for every outcome (Table 6) and collapse each domain into a single standardised index (the mean of its components' control-standardised z-scores) so that the number of outcomes does not inflate the number of tests (Kling, Liebman and Katz, 2007); the index ITTs, with randomisation-inference p-values, are in Table 7.

Table 6
Intention-to-treat effects by outcome

Outcome	Control mean	ITT (clustered SE)
Panel A: Healthcare use		
Prescription drugs (count)	2.32	0.100** (0.051)
Doctor visits (count)	1.91	0.314*** (0.054)
ER visits (count)	0.47	0.007 (0.016)
Hospital admissions (count)	0.10	0.006 (0.006)
Panel B: Financial strain (higher = more strain)		
Any out-of-pocket medical cost	0.55	−0.058*** (0.008)
Any medical debt owed	0.60	−0.052*** (0.008)
Borrowed / skipped bills to pay for care	0.36	−0.045*** (0.007)
Refused care over unpaid bills	0.08	−0.011** (0.004)
Panel C: Health		
Good/very-good health	0.55	0.039*** (0.008)
Health not poor	0.86	0.029*** (0.005)
Health improved (vs last year)	0.71	0.033*** (0.007)
Days not in poor health, total (/30)	21.9	0.381** (0.162)
Days not in poor physical health (/30)	20.3	0.459*** (0.175)
Days not in poor mental health (/30)	18.7	0.603*** (0.184)
Screens negative for depression	0.67	0.023*** (0.007)

Table 7
ITT on standardised domain indices, with randomisation inference

Domain index	ITT (SE), control-SD units	RI p-value (2,000 permutations)
Healthcare use	0.045*** (0.011)	0.0005
Financial strain	−0.089*** (0.010)	0.0005
Health	0.062*** (0.011)	0.0005

Note: Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sign orientation: the financial-strain items keep their natural direction, so higher = more strain; a negative ITT on that index is therefore an improvement (less out-of-pocket

spending, debt, and borrowing). Randomisation-inference p-values are based on 2,000 placebo re-draws of the lottery at the household level within household-size strata.

The offer raises healthcare use (driven by doctor visits and prescriptions, not ER or hospital use), sharply reduces financial strain (the single most robust result, with every component improving), and improves self-reported and, especially, mental health. The randomisation-inference p-values (≈ 0.0005 : no placebo reshuffle of the lottery reproduces an effect as large as the observed one) confirm these conclusions without leaning on the normal approximation. We are deliberately honest about the limit of the health result: the OHIE in-person exam found no significant change in objective physical biomarkers: blood pressure, cholesterol, glycated haemoglobin (Baicker et al., 2013). “Health improved” here means perceived and mental health, not measured physical disease.

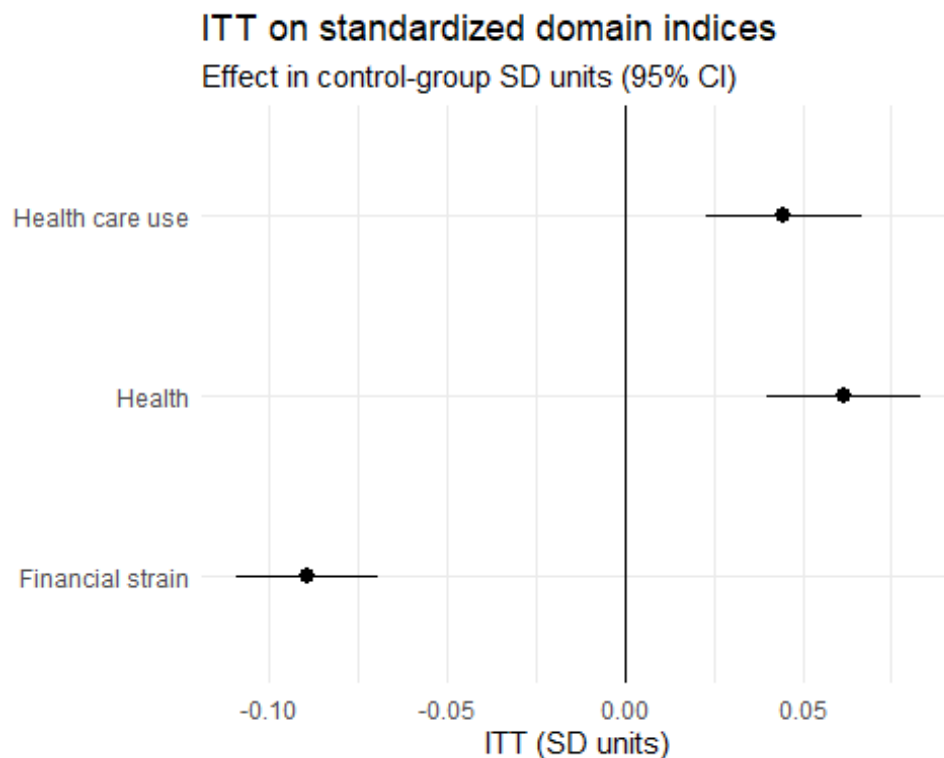


Figure 7. ITT on the three standardised domain indices, in control-group SD units, with 95% confidence intervals. A confidence interval that excludes zero corresponds to significance at the 5% level.

The same intention-to-treat effects are shown outcome by outcome in Figure 8, which makes the within-domain pattern visible: the healthcare-use effect is concentrated in doctor visits and prescriptions, financial strain falls on every component, and the health gains load on the self-reported and mental-health measures.

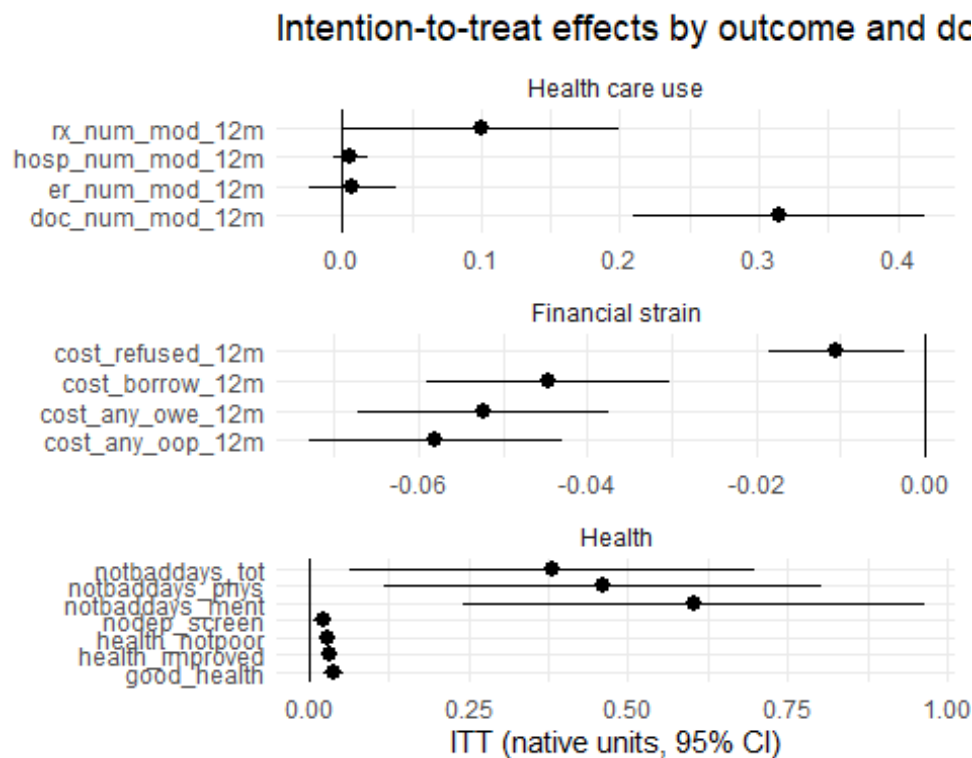


Figure 8. Intention-to-treat effects outcome by outcome, in control-group SD units, with 95% confidence intervals. This disaggregates the three standardised domain indices of Figure 7 into their individual components.

3.3 Heterogeneity by age

We test whether effects differ for older adults (50–64), who carry more chronic disease, by interacting the offer with an older-age indicator (Table 8).

Table 8
Heterogeneous ITT by age (standardised indices, SD units)

Domain index	ITT, younger (19–49)	ITT, older (50–64)	Difference	p (interaction)
Healthcare use	0.031	0.073	+0.042	0.070
Financial strain	–0.088	–0.092	–0.001	0.968
Health	0.052	0.085	+0.033	0.143

Note. the difference is the coefficient on the offer $\times$ older interaction; household-clustered SEs. None of the differences is significant at the 5% level (healthcare use is marginal at the 10% level).

Point estimates are larger for older adults, most visibly for healthcare use (marginally significant at 10%), while financial protection is essentially age-invariant. We do not over-read this: splitting the sample lowers power, so a non-significant interaction is not evidence of homogeneity.

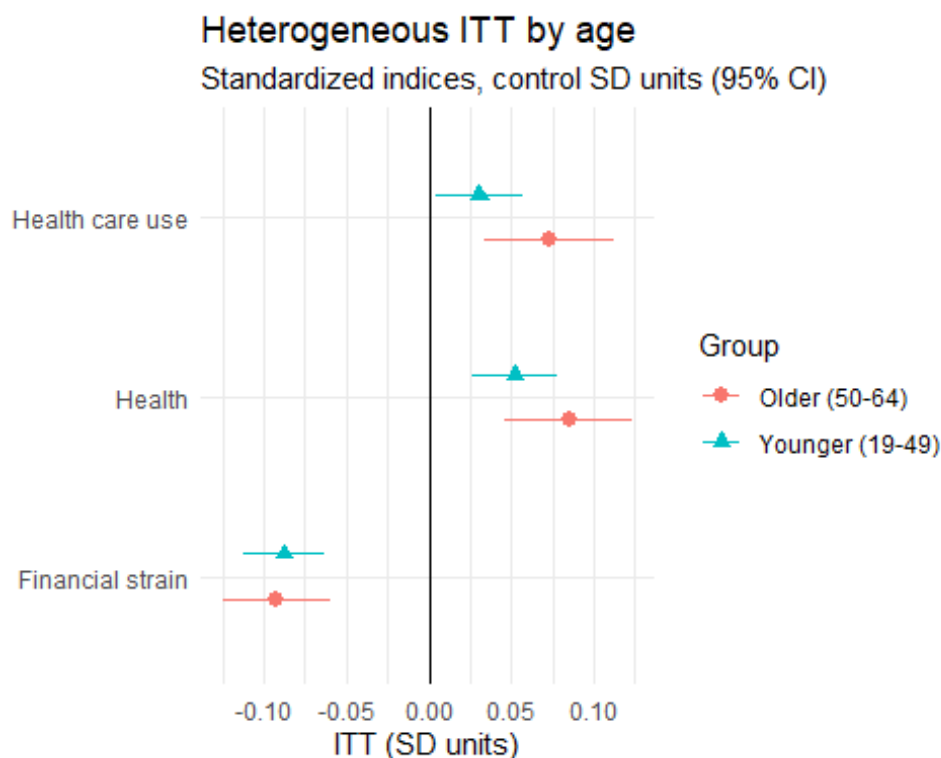


Figure 9. Heterogeneous intention-to-treat effects by age group (younger 19–49 versus older 50–64) on the three standardised domain indices, with 95% confidence intervals.

3.4 The shared outcome: mortality

Mortality is administrative and observed for the full lottery sample, so we estimate it without survey weights, on survival (alive = 1 – death) with draw and household-size fixed effects and clustered SEs. The ITT on survival is +0.0003 (clustered SE 0.0007, $p = 0.64$), a precise null, not evidence of no effect. Control-arm mortality is only $\approx 0.85\%$, so the experiment is badly underpowered for so rare an event over a ≈ 16 -month horizon: the confidence interval, re-expressed against control mortality, cannot rule out anything from roughly a 20% reduction to a modest increase in deaths. This mirrors Finkelstein et al. (2012); experimental designs with greater statistical power have since detected mortality reductions from coverage (Goldin, Lurie and McCubbin, 2021). It is the cleanest bridge to Part I, the exact outcome a state-level cross-section could not pin down, now estimated under randomisation but limited by power rather than by identification.

4. Interpretation and Critical Discussion

The substantive verdict is real but uneven, and the honest reading matters more than a headline. Because winning raised enrolment by ≈ 25 pp, the experiment is informative; coverage clearly increased healthcare use and, most robustly, reduced financial strain, while improving self-reported and mental health. It did not move physical biomarkers within a year, and mortality is a precise null, outcomes that are underpowered, not zero.

The methodological lesson is the part most worth carrying away, and it closes the arc with Part I. On identical data, estimating the effect of enrolment by OLS or by matching returns coefficients near zero, far from the experimental LATE; matching balanced every observable and still failed, because selection ran on unobservables. Adjustment is not identification. Part I could only hypothesise this confounding from an aggregate cross-section; Part II made it measurable and then removed it by design.

On the credibility of the assumptions: balance and the strong first stage support random assignment and relevance; exclusion and monotonicity are plausible given that the lottery altered nothing but eligibility to apply; randomisation inference corroborates the asymptotic clustered-SE inference. The honest limitations are: (i) external validity: one state, one period, a short horizon, so generalisation to other populations and to the long run is not guaranteed; (ii) imperfect take-up, which is exactly why ITT and LATE are distinct objects and are kept separate; (iii) differential non-response (≈ 1.7 pp), which can reintroduce selection among respondents despite weighting; and (iv) low subgroup power, which forbids strong heterogeneity claims.

Conclusion

This paper asked one question: does health insurance causally improve health and financial well-being? It answered that question with two designs that, taken together, show how much the answer depends on the way it is identified. Part I regressed state-level mortality on insurance coverage and could not recover a credible causal effect: the association is weak, unstable across specifications, and changes sign once income is added. We argued that this is not a “not enough controls” problem but a structural one, driven by omitted-variable bias, reverse causality, and the ecological fallacy. Part I therefore works as a diagnosis. Part II provided the cure: the 2008 Oregon lottery randomly decided who could apply for Medicaid, so a comparison of winners and losers identifies the causal effect of the offer of coverage without relying on any of the assumptions that fail in Part I.

The substantive results are real but uneven, and the honest reading matters more than a single headline number. Winning the lottery raised Medicaid enrolment by about 25 percentage points, which is what makes the experiment informative. Coverage clearly increased healthcare use and, most robustly, reduced financial strain; it also improved self-reported and, in particular, mental health. What it did not do, within the one-year horizon, is move objective physical biomarkers (Baicker et al., 2013), and we cannot detect any effect on mortality. These are precise nulls, not evidence of no effect: mortality in the control group is only about 0.85%, so the experiment is underpowered for such a rare outcome over so short a window. “Underpowered” and “zero” are not the same statement.

The methodological lesson is what we would most want a reader to take away. On the very same OHIE sample, estimating the effect of enrolment by naive OLS or by propensity-score matching returns coefficients close to zero, far from the experimental LATE. Matching balances every observable covariate and still gives the wrong answer, because enrolment is selected on unobservables that no observational adjustment can see. This is exactly the pathology of Part I, reproduced in miniature inside Part II. The practical conclusion is blunt: controlling for more variables does not buy identification; only a credible source of exogenous variation can separate the causal effect from the selection.

These conclusions come with limitations we do not hide: the experiment is local and short-run, take-up was imperfect (so ITT and LATE must be kept distinct), survey response was slightly

lower among winners, and the age-subgroup results are too imprecise for strong heterogeneity claims. The most natural extension points at the time dimension. A single lottery identifies a local, short-run effect; the population-wide, longer-run effect of expanding Medicaid is better studied through the staggered ACA expansions adopted by different states from 2014 onward.

That is a staggered-adoption setting, where the conventional two-way fixed-effects estimator is generally biased under heterogeneous or dynamic treatment effects because it uses already-treated units as controls (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020). Modern estimators were built to fix this: group-time effects aggregated into event studies (Callaway and Sant'Anna, 2021), robust event-study estimation (Sun and Abraham, 2021), and efficient imputation (Borusyak, Jaravel and Spiess, 2024), with Roth, Sant'Anna, Bilinski and Poe (2023) providing a synthesis.

In short, our experimental result answers “does it work, here and now?”, while “does it keep working, everywhere, over time?” is exactly what this newer difference-in-differences literature is designed to answer.

Acknowledgments

We are grateful to Professor Roberta Di Stefano for her support and guidance throughout this group project. Her lectures on causal inference provided the conceptual framework for the paper, and her advice on the identification strategy was invaluable in moving our analysis from correlation toward credible causal evidence. We also thank Professor Nina Deliu, who co-taught the course, for the statistical-inference foundations underlying Part I. Any remaining errors are our own.

Use of AI Tools

In accordance with the course policy, we disclose that AI tools were used solely as a support for coding (debugging and refactoring R scripts), for editing and language polishing of the written report, and for formatting tables. All research design choices, model specifications, empirical results, and their interpretation are the authors' own. No analytical result was generated or reported without being produced and verified directly in R.

Code and Data Availability

All replication materials are available in a public GitHub repository at https://github.com/MatteoSep/Data-Hunters_SI-Group-Project: the R scripts that produce every result, the R Markdown source of this report, the compiled paper, the datasets, and the code for the planned difference-in-differences extension.

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